

Moaz Mustafa

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EDUCATION

Vellore Institute of Technology
B.Tech. in Computer Science and Engineering

Expected May 2027
CGPA: 8.34/10.0

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, GDScript, Java, C, HTML, CSS
Frameworks: React, Next.js, FastAPI, Node.js, Godot 4, Tailwind CSS, Tauri
Systems and Infrastructure: Linux, Wayland, Hyprland, PostgreSQL, Supabase, Docker, Redis, Apache Kafka
Tools and Practices: Git, GitHub Actions, Vitest, Playwright, pytest, REST APIs, CI/CD

EXPERIENCE

Lahmah Cuts
Software Engineering Intern

May 2026 – July 2026
Hybrid, Bahrain

- Engineered a **production-grade commerce platform** with **Next.js, React, TypeScript, Supabase, and PostgreSQL**, delivering secure authentication, checkout, order management, and **role-based administrative workflows**.
- Designed reliable **payment, receipt, and refund pipelines** using verified webhooks, **idempotent transactions**, reconciliation jobs, immutable financial records, and automated **email and WhatsApp delivery**.
- Strengthened release quality with **PostgreSQL Row-Level Security**, role-based access controls, Dockerized testing, and **GitHub Actions CI/CD** covering builds, API smoke tests, and **more than 260 automated test cases**.

PROJECTS

Hyperspace

Rust, Tokio, Godot 4, GDScript, Hyprland, Wayland, Unix Sockets

- Developing an experimental **3D desktop shell for Arch Linux and Omarchy** that represents applications, workspaces, projects, and **real-time system telemetry** as interactive spatial objects while preserving **Hyprland as the underlying Wayland compositor**.
- Architected a **Rust and Tokio systems daemon** for **XDG application discovery, process control, Hyprland IPC**, and Linux telemetry, communicating with the Godot frontend through **Unix domain sockets and a typed JSON protocol**.

mdForge

Rust, TypeScript, Tauri, Tera, Vite, React, HTML/CSS, GitHub Actions

- Reduced binary size **10–20× vs Electron** and RAM from **~500 MB to ~50 MB** by architecting a **concurrent Rust + Tauri backend** with optimized I/O throughput, achieving **sub-100 ms live preview latency** at hundreds of files per second.
- Shipped reliable **cross-platform releases for Windows, macOS, and Linux** by implementing **GitHub Actions CI/CD** with a code review-gated PR workflow, enabling confident iterative delivery across all three targets.

Topper-Inator

Python, FastAPI, TypeScript, React, Tailwind CSS, Next.js, Supabase, Vercel, Render

- Reduced YouTube playlist ingestion latency to **1.37 s for 6 videos (4.38 videos/sec)** by designing a **FastAPI REST API** with async batch workflows, **p50/p95 telemetry**, and success-rate reporting that guided iterative performance optimization.
- Cut per-video transcript extraction to **11.7 s** by designing the system end-to-end across a **Next.js frontend, FastAPI backend, and Supabase data layer** — collaborating with a teammate on frontend integration and conducting code reviews before each production deployment.